

PROBLEM SET 1—SOLUTIONS

ECON301—ECONOMETRICS I

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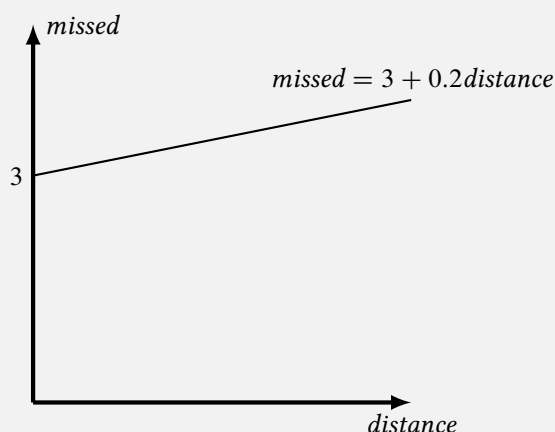
Instructions: Due on 07/02/2020 before 12 PM in the envelope pasted to my office door. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission. Don't forget to attach the cover sheet.

1. Do problems 2 and 11 from appendix A from the book.

Solution

Problem 2.

- (i) Below is the graph. The point (0, 3) is the vertical intercept. The slope is 0.2. The slope means that for every increase in 1 mile from school, the average number of classes missed increases by 0.2. Perhaps a more intuitive interpretation is as follows: for every 10 extra miles away from school, the average number of missed classes increases by 2.



- (ii) $3 + 0.2(5) = 4$ classes
- (iii) The difference in the average number of classes missed for someone who lives 10 miles away and someone who lives 20 miles away is $10(.2) = 2$ classes.

Problem 11.

- (i) Consider points (x_1, y_1) and (x_2, y_2) . Since they are on the line $y = \beta_0 + \beta_1 x$ then the two points satisfy this equation, that is

$$y_i = \beta_0 + \beta_1 x_i, \quad i = 1, 2.$$

We need to show that the point $(\bar{x}, \bar{y})$ also satisfies the equation of the line. Adding up both equations yields:

$$\sum_{i=1}^2 y_i = 2\beta_0 + \beta_1 \sum_{i=1}^2 x_i,$$

then we divide the whole equation by 2 to obtain:

$$\bar{y} = \beta_0 + \beta_1 \bar{x},$$

which shows that $(\bar{x}, \bar{y})$ is also on the line.

(ii) Analogous to the previous case.

2. Probability review. Do problems 2, 4, 5, 6, and 10 from appendix B from the book.

Solution

Problem 2.

- (i) $P(X \leq 6) = P[(X-5)/2 \leq (6-5)/2] = P(Z \leq .5) = P(Z \leq 0.5) \approx 0.692$, where Z denotes a Normal $(0, 1)$ random variable.
- (ii) $P(X > 4) = P[(X-5)/2 > (4-5)/2] = P(Z > -.5) = P(Z \leq .5) \approx .692$.
- (iii) $P(|X-5| > 1) = P(X-5 > 1) + P(X-5 < -1) = P(X > 6) + P(X < 4) \approx (1-.692) + (1-.692) = .616$, where we have used the answers from the previous two items.

Problem 4. We want $P(X \geq 0.6)$. Because X is continuous, this is the same as $P(X > 0.6) = 1 - P(X \leq 0.6) = F(0.6) = 3(0.6)^2 - 2(0.6)^3 = 0.648$. One interpretation of this result is that almost 65% of all counties have an elderly employment rate of 0.6 or higher.

Problem 5.

- (i) If X is the number of jurors convinced of Simpson's innocence, then $X \sim \text{Bin}(12, 0.20)$. We need to find $P(X \geq 1) = 1 - P(X = 0) = 1 - (0.8)^{12} \approx 0.931$.
- (ii) Above, we computed $P(X = 0)$ as about .069. We need $P(X = 1)$, which we obtain from the pdf of the binomial $\binom{n}{x} p^x (1-p)^{n-x}$ with $n = 12$, $p = .2$, and $x = 1$: $P(X = 1) = 12 \cdot (.2)(.8)^{11} \approx .206$. Therefore, $P(X \geq 2) \approx 1 - (.069 + .206) = .725$, so there is almost a three in four chance that the jury had at least two members convinced of Simpson's innocence prior to the trial.

Problem 6.

$$E(X) = \int_0^3 x f(x) dx = \int_0^3 x (1/9) x^2 dx = (1/9) \int_0^3 x^3 dx = (1/9) (1/4) x^4 \Big|_0^3 = 9/4 = 2.25.$$

Thus, the expected prison sentence is 2.25 years

Problem 10.

- (i) $E(\text{GPA} | \text{SAT} = 800) = .70 + .002(800) = 2.3$. Similarly, $E(\text{GPA} | \text{SAT} = 1,400) = .70 + .002(1400) = 3.5$. The difference in expected GPAs is substantial, but the difference in SAT scores is also rather large.
- (ii) Following the hint, we use the law of iterated expectations. Since $E(\text{GPA} | \text{SAT}) = .70 + .002 \text{SAT}$, the (unconditional) expected value of GPA is $.70 + .002 E(\text{SAT}) = .70 + .002(1100) = 2.9$.
- (iii) If a student's SAT score is 1,100, this does not mean he or she will have the GPA 2.9, found in part(ii). The GPA of 2.9 is the average GPA calculated when the average SAT score is 1,100. GPA is influenced by other variables.

3. **Statistics review.** Do problems 1, 2, and 4 from appendix C from the book.

Solution

Problem 1. For this problem, recall that the four observations are independent, and identically distributed. The observations are drawn from a population with mean and variance μ and σ^2 respectively.

- (i) $E[\bar{Y}] = \mu$ and $V[\bar{Y}] = \sigma^2/4$.
- (ii) $E[W] = \mu \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{4} + \frac{1}{2} \right) = \mu$ and $V[W] = \sigma^2 \left(\frac{1}{64} + \frac{1}{64} + \frac{1}{16} + \frac{1}{4} \right) = \frac{11}{32}\sigma^2$. Notice that since $E[W] = \mu$ the estimator is unbiased.
- (iii) Both $\bar{Y}$ and W are unbiased. However, $\bar{Y}$ has a lower variance than W , thus the former is more efficient. Hence, $\bar{Y}$ is preferred.

Problem 2.

- (i) $E(W_a) = \sum_{i=1}^n a_i E(Y_i) = m \left(\sum_{i=1}^n a_i \right)$. Therefore, we must have $\sum_{i=1}^n a_i = 1$ for unbiasedness.
- (ii) $V(W_a) = \sum_{i=1}^n a_i^2 V(Y_i) = \sigma^2 \left(\sum_{i=1}^n a_i \right)$.
- (iii) Whenever $\sum_{i=1}^n a_i = 1$ —which is the condition for unbiasedness—then we have that $\frac{1}{n} \leq \sum_{i=1}^n a_i^2$. Note that this inequality is a corollary of the famous Cauchy-Schwarz inequality.^a But then

$$V(\bar{Y}) = \frac{\sigma^2}{n} \leq \sigma^2 \left(\sum_{i=1}^n a_i \right) = V(W_a).$$

Problem 4.

- (i) Using the hint, $E(Z|X) = E(Y/X|X) = E(Y|X)/X = \theta X/X = \theta$. It follows from the law of iterated expectations, that $E(Z) = E(\theta) = \theta$.
- (ii) This follows from part (i) and the fact that the sample average is unbiased for the population average. Write

$$W_1 = \frac{1}{n} \sum_{i=1}^n \frac{Y_i}{X_i} = \frac{1}{n} \sum_{i=1}^n Z_i$$

where $Z_i = Y_i/X_i$. From part (i) we get that $E(Z_i) = \theta$ for all i .

- (iii) In general, the average ratios, Y_i/X_i , is not the ratio of the averages $W_2 = \bar{Y}/\bar{X}$. Nevertheless, W_2 is also unbiased. We will use the law of iterated expectations to show this. First, note that $E(Y_i|X_1, \dots, X_n) = E(Y_i|X_i)$ under random sampling because the observations are independent. Therefore, $E(Y_i|X_1, \dots, X_n) = \theta X_i$ and so

$$\begin{aligned} E(\bar{Y}|X_1, \dots, X_n) &= \sum_{i=1}^n \frac{1}{n} E(Y_i|X_1, \dots, X_n) = \sum_{i=1}^n \frac{1}{n} E(Y_i|X_i) = \sum_{i=1}^n \frac{1}{n} \theta X_i \\ &= \frac{\theta}{n} \sum_{i=1}^n X_i = \theta \bar{X}. \end{aligned}$$

Therefore, $E(W_2|X_1, \dots, X_n) = E(\bar{Y}/\bar{X}) = \theta \bar{X}/\bar{X} = \theta$, which means that W_2 is unbiased when conditioned on $X_1, \dots, X_n$. Another application of the law of iterated expectations yields that W_2 is unbiased.

- (iv) For the $n = 17$ observations given in the table, the point estimates are $w_1 = 0.418$ and $w_2 = 129.09/308.76 = .418$. The estimates for θ are similar in this example. Notice that θ can be interpreted as the “corn yield per corn pixel.” Thus, each corn pixel is estimated to yield 0.418 hectares of corn. For example, if we have $x = 300$, i.e., 300 pixels of corn, Then the predicted yield is $0.418(300) = 125.4$ hectares.

^aFor any vectors u and v in $\mathbb{R}^n$, then $|u \cdot v|^2 \leq \|u\|^2 \|v\|^2$ where $\cdot$ denotes the dot product and $\|x\|$ denotes the norm of the vector x .